The Ramification Index:

Version 3: Provenance, Construction, and What the Series Can Support

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Abstract

This third version of the Ramification Index is a correction and a method, not a new result. It reports four things. (1) The primary source underlying the 1980–2017 segment of the index—the McCallum memory-price archive—is no longer available at its published address; the domain has lapsed and now serves unrelated content. The original workbook has been recovered from the Internet Archive and is republished here, restoring the reproducibility that Version 2 claimed. (2) The 1988 "RAM drought", one of the two largest supply shocks in the series' history, is absent from the published price table: the recovered archive shows a +82.5% annual move where the published series shows −21.8%, and Version 1's own event table documents the spike that its price table omits. This bears directly on the claim that the 2025–2026 episode is without historical precedent. (3) A point-in-time construction standard is specified and implemented, together with a monthly index (RI-M) built from the recovered archive. RI-M satisfies the standard by proof but fails two of the protocol's own guards as a measurement instrument, and its null is therefore reported as *untestable* rather than as a rejection. (4) The Version 2 Granger result is carried by its last six observations: withholding 2020–2026 moves the forward p-value from 0.0243 to 0.2984 and flips the contemporaneous control from survival to death, and the published annual panel ($n = 46$) cannot in any case detect a partial correlation below 0.508 at a multiple-comparison-corrected threshold. The result is accordingly restated as untested pending higher-frequency data rather than as established. A corrected quality-adjusted series is published alongside the original—nothing is overwritten—together with an eight-guard test protocol and its implementation.

Keywords: economic indicators; semiconductors; DRAM; Ramification Index; reproducibility; Granger causality; statistical power; point-in-time data.

1 Introduction

Version 2 of this paper argued that the average selling price of DRAM leads the United States business cycle, and supported that argument with a bivariate vector autoregression

on annual and quarterly data. This version does not extend that argument. It examines whether the data and specification underlying it can carry the weight placed on them, and concludes that in several respects they cannot yet.

The motivation was methodological. A companion study applied a battery of lead-lag controls to the agricultural input complex and found that four apparently publishable results were artifacts, each surviving every check in place at the time and dying only when a further check was added [Wakil 2026a]. None was a coding error. Each was a defensible-looking specification choice. Applying the same battery here was intended as due diligence on this paper's own claims, and it surfaced problems in the data before it surfaced any in the econometrics.

Section 2 reports the loss and recovery of the primary source. Section 3 corrects the description of how the index is constructed. Section 4 documents the missing 1988 episode. Section 5 specifies a point-in-time construction standard and applies it. Section 7 reports the monthly index and why it does not work. Section 8 gives the power analysis. Section 9 states the protocol. Section 10 states a scope condition for the class of indicator to which the Ramification Index belongs. Section 11 lists what is withdrawn and what is restated.

2 Provenance: a source that stopped existing

Version 2, Section 2.2, is titled "It has a published, reproducible time series going back to 1957." That sentence is no longer true.

The series in question is John C. McCallum's memory-price archive, the de facto standard reference for DRAM price per unit capacity, reproduced in Hennessy and Patterson's *Computer Architecture* and used in AI Impacts' semiconductor trend work [McCallum, Hennessy & Patterson, AI Impacts]. It was published at `jcmit.net/memoryprice.htm`. That domain has lapsed and now resolves to an unrelated commercial site. A reader following the citation in Version 1 or Version 2 today reaches nothing.

The underlying workbook was recovered from the Internet Archive at `https://web.archive.org/web/20201029015902id_/http://www.jcmit.net/MemDiskPrice-xl95.xls`. Its OLE document metadata records *Author: John McCallum*, created 1998-11-09, last saved 2020-09-20. The MEMORY sheet contains 380 dated observations from 1957 to 2020-09, of which 342 fall on a monthly grid from 1985-01 onward. It is redistributed with this paper, and Figure 1 plots it in full, with its provenance stated on the face of the figure — the practice this section argues for.

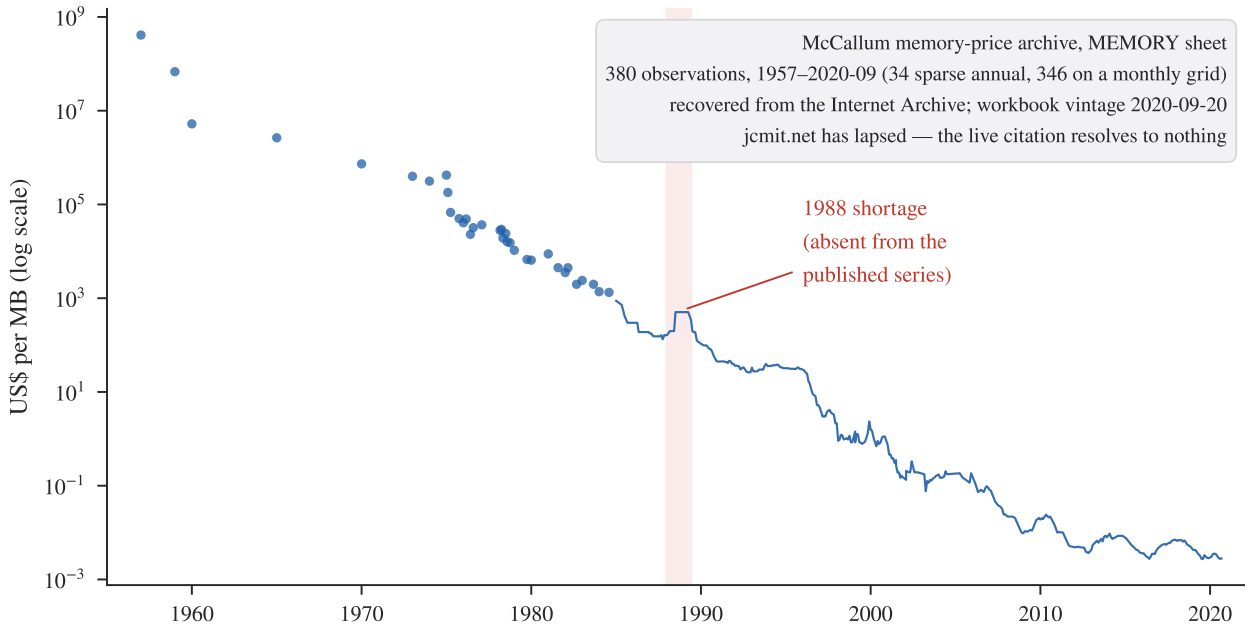


Figure 1: The recovered McCallum archive, 1957–2020, in US$ per megabyte on a log scale. Sparse annual observations (dots) give way to a monthly grid from 1985. The 1988 episode — the subject of Section 4 — is visible on the face of the raw data and absent from the published price table. Generated by `analysis/make_figures_v3.py` from the vendored workbook.

Remark 2.1. The failure mode is worth naming because it is general. A citation to a personal academic website is a citation to a resource with no institutional commitment to persistence. The archive survived only because a third party mirrored it. Any index built on such a source should vendor its inputs at the version used, which is the practice adopted in Section 5.

3 What the index is actually made of

Version 2's data section describes the price series as "McCallum archive (1980–2017) chain-linked to TrendForce DDR4 contract ASP (2018–2026)."

That description is incorrect, and Version 1 states the correct one. Version 1's research documentation describes the series as "mid-year average retail for a typical mainstream module, *rounded*", with sources given as "McCallum (jcmit), hblok.net, TrendForce", and notes that pre-1995 figures track the then-current DRAM generation while 1995+ figures are SIMM/DIMM retail.

The published series is therefore a rounded three-source blend of retail module prices with a generational composition break in the mid-1990s, not a chain-link of two series. Three consequences follow.

1. **It does not reproduce from any single source.** Aggregating the recovered archive to annual frequency and comparing log-differences against the published series gives

$r = 0.597$. Six aggregation rules were tried—mean of monthly medians, mean of all observations, median, minimum, first, and last—and the best, minimum-of-year, reaches 0.737. This is a consequence of blending, not evidence of an error, but it means the series cannot be independently regenerated from what is cited.

2. **Rounding enters the index directly.** The index is $R_t = \Delta \log \mathrm{ASP}_t$. Values such as 107, 73 and 0.0030 carry two to three significant figures, so quantization noise propagates into the quantity being tested rather than into a level that is only read off a chart.

3. **The composition break is undocumented in the index.** A series that tracks "the then-current generation" before 1995 and mainstream DIMM retail afterwards is measuring different goods on either side of the join, and no test in Version 2 conditions on that.

4 The 1988 RAM drought is not in the series

This is not a documentation issue.

Version 1's event table records 1988 as "The Year of the RAM Drought", a post-dumping capacity underinvestment episode, with the annotation "Prices spiked ∼60% YoY" [Tedium]. Version 1's price table, in the same document, records 1988 at \$107/MB against \$133 in 1987—a decline of 21.8% in logs. The two tables contradict each other, and it is the price table that propagates into `indices-wide.csv` and into every estimate in Versions 1 and 2.

The recovered archive settles which one is right.

Year	Published \$/MB	Published $\Delta \log$	Archive \$/MB	Archive $\Delta \log$
1987	133	−29.1%	158	−37.2%
1988	**107**	**−21.8%**	**361**	**+82.5%**
1989	73	−38.2%	300	−18.6%

Table 1: The 1988 shortage in the published series and in the recovered archive. Archive figures are annual means of monthly observations.

The decisive evidence is a single product line. JDR's "MCT-ATMF 3MB Multifunction" board is quoted continuously through the episode, and the archive itemises the quote: \$169.90 for the board plus 36 DRAM chips. The chip price runs \$2.95 in April 1987, \$3.49 in February 1988, and \$12.45 by July 1988. Same vendor, same board, same capacity—the memory alone rises by a factor of 3.6. Whatever else is true of the archive, this is not a module-composition artifact and not an aggregation choice. Both readings are plotted in Figure 2: the left panel shows the published table declining smoothly through an episode in which the archive nearly triples, and the right panel shows the same-board evidence that settles which is right.

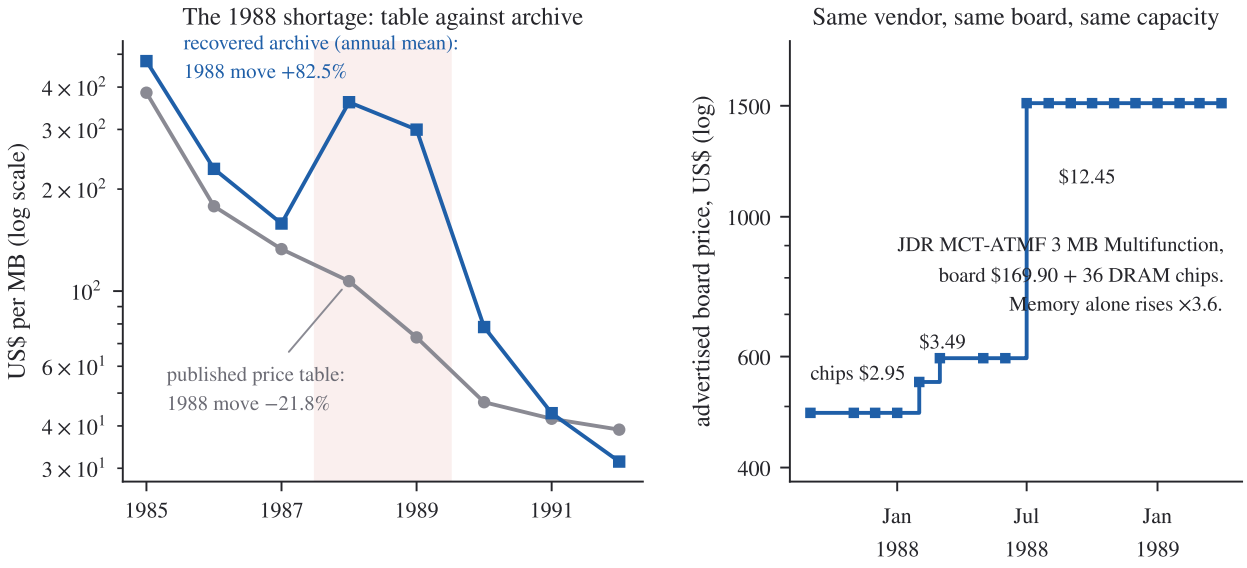


Figure 2: The 1988 shortage. *Left:* the published price table against the recovered archive (annual means of monthly observations), 1985–1992. The published series declines 21.8% in logs through 1988; the archive rises 82.5%. *Right:* the JDR MCT-ATMF 3 MB Multifunction board, quoted continuously through the episode — same vendor, same board, same capacity — with the itemised DRAM chip price at each step. Generated by `analysis/make_figures_v3.py`.

Three independent readings of the archive agree with Version 1's narrative and against its price table:

Reading of 1988	Δ log
Raw archive, annual mean	+82.5%
Hedonic, quality-adjusted (§6)	+70.0%
Version 1 narrative table	≈ +60%
Version 1 price table (the one that ships)	−21.8%

Two implications.

First, the recession-coverage and correlation results in Versions 1 and 2 are computed on a series that omits a major supply shock. The direction of the resulting bias is not obvious and is not estimated here.

Second, and more directly: Version 2's central new claim is that the 2025–2026 episode constitutes a supply-side divergence regime "without historical precedent" in the 46-year series. The nearest historical precedent is a supply-driven price spike caused by capacity underinvestment following a trade intervention. The series as published does not contain it. The uniqueness claim cannot be evaluated on this data, and is withdrawn pending correction (Section 11).

Remark 4.1. Comparing the Version 1 and Version 2 panels directly, exactly one row differs: 2026, revised from \$0.0052/MB to \$0.0097/MB, an increase of 86.9%. Every other year is identical. The final row is a live forecast standing in a column of realised history, and it moved by a factor approaching two between versions. Any estimate including it conditions realised-history coefficients on projected data.

5 A point-in-time construction standard

The companion agricultural study's most expensive error was look-ahead bias baked into the *construction of a signal* rather than into the test applied to it: a confound regression fit on the full sample made the signal at month t depend on coefficients estimated through the end of the sample. Correcting it cost half the result. The error is invisible downstream, because the test itself is specified correctly.

Definition 5.1 (Point-in-time index)**.** An index $\{R_t\}$ is *point-in-time* if for every t, the value R_t computed from data available up to and including t is identical to the value computed from the complete sample.

This is a testable property, not a description of intent. The test is to rebuild the index on truncated inputs and compare the overlap.

Version 2's index cannot currently be tested for it, because two of its inputs are unpublished: the ratio used to chain-link the 2018 segment, and the `hbm_weight` series used to construct the bifurcated sub-indices. If either was computed with knowledge of how the sample resolved, every historical value of the composite embeds that knowledge.

Three further requirements are adopted here, each because omitting it caused a documented failure elsewhere:

1. **No forward-filling across gaps.** A missing month must produce no return, not a zero. A fabricated zero is indistinguishable from a market that did not move, which is the condition Guard 6 (Section 9) exists to detect.

2. **No smoothing or seasonal adjustment at construction.** Transformations belong in the test, where they can be run both ways and compared, not in the series where they cannot be undone.

3. **Vintage stamping.** Source, source version, and retrieval date travel with the data, so that a figure can be traced to the version of the input that produced it—the operation that became impossible for Version 2 once `jcmit.net` lapsed.

6 Correcting the series

The archive cannot simply be substituted for the published series, because its raw \$/MB figures are contaminated by module composition. The 1989–90 segment walks from 3 MB JDR boards to 4 MB Unitex, then 8 MB Bocaram, then 1 MB SIMMs. A board's fixed cost

amortises over its capacity, so part of the apparent decline is the container changing rather than the memory getting cheaper. Differencing that series measures both.

Two standard repairs exist and only one works here.

Matched-model—take returns only between consecutive quotes of the same product—is clean, and yields 76 usable pairs in 35 years implying a long-run decline of -5.4%/yr against a documented $\approx -36\%$/yr. Nearly all DRAM deflation is realised *at* product replacement, which is precisely the link a matched-model index discards. This is textbook new-goods bias, and it is why the BLS runs hedonic regressions on electronics rather than matched models.

Hedonic—regress log price per MB on log capacity and a year effect—controls the composition directly:

$$\log\left(\tfrac{\$}{\text{MB}}\right)_i = \alpha + \beta \log(\text{capacity}_i) + \sum_t \gamma_t \mathbb{1}[\text{year}_i = t] + \varepsilon_i$$

with the quality-adjusted index read off $\{\gamma_t\}$. On 346 observations, $R^2 = 0.9955$ and $\hat{\beta} = -0.109$ (s.e. 0.037, $t = -3.0$): larger modules do cost less per megabyte, significantly so. The resulting index declines 39.2%/yr over 1985–1999 and 10.7%/yr from 2010, against published benchmarks of $\approx 36\%$/yr long-run and $\approx 15\%$/yr post-2010 [AI Impacts]. The matched-model alternative misses both by a wide margin.

Remark 6.1 (Guard 4 applied to the correction itself)**.** A hedonic fit on the whole sample makes the 1988 index value depend on observations from 2020. The expanding-window estimator refits for each year t using only data dated t or earlier. The two agree to a correlation of 1.000, with mean absolute difference 0.9pp and a maximum of 5.0pp in 1999. Leakage is negligible here—the opposite of the agricultural case, where the same check cost half the result. It was worth running precisely because that is not knowable in advance. The point-in-time variant cannot reach back to 1988, since it requires a training window; the published correction therefore uses the full-sample fit and reports this agreement as its warrant.

Remark 6.2 (What the correction does not fix)**.** Quality adjustment does not rescue Guard 7. Against the import price index the correlation moves from 0.572 to 0.567; against the export index, 0.364 to 0.352; against the semiconductor PPI, 0.455 to 0.463. The residual disagreement is a difference in price *concept*—advertised retail quotes against transaction-price surveys—not a difference in quality mix, and no capacity control can close it. The corrected series is better on its own terms: it reproduces the deflation benchmarks, its capacity coefficient is correctly signed and significant, and it contains 1988. It is not thereby a survey-grade price index, and is published as an auditable alternative rather than as a drop-in replacement. Both series ship in `data/indices-wide-v3.csv` with a per-year provenance column; nothing was overwritten. Figure 3 shows the two side by side.

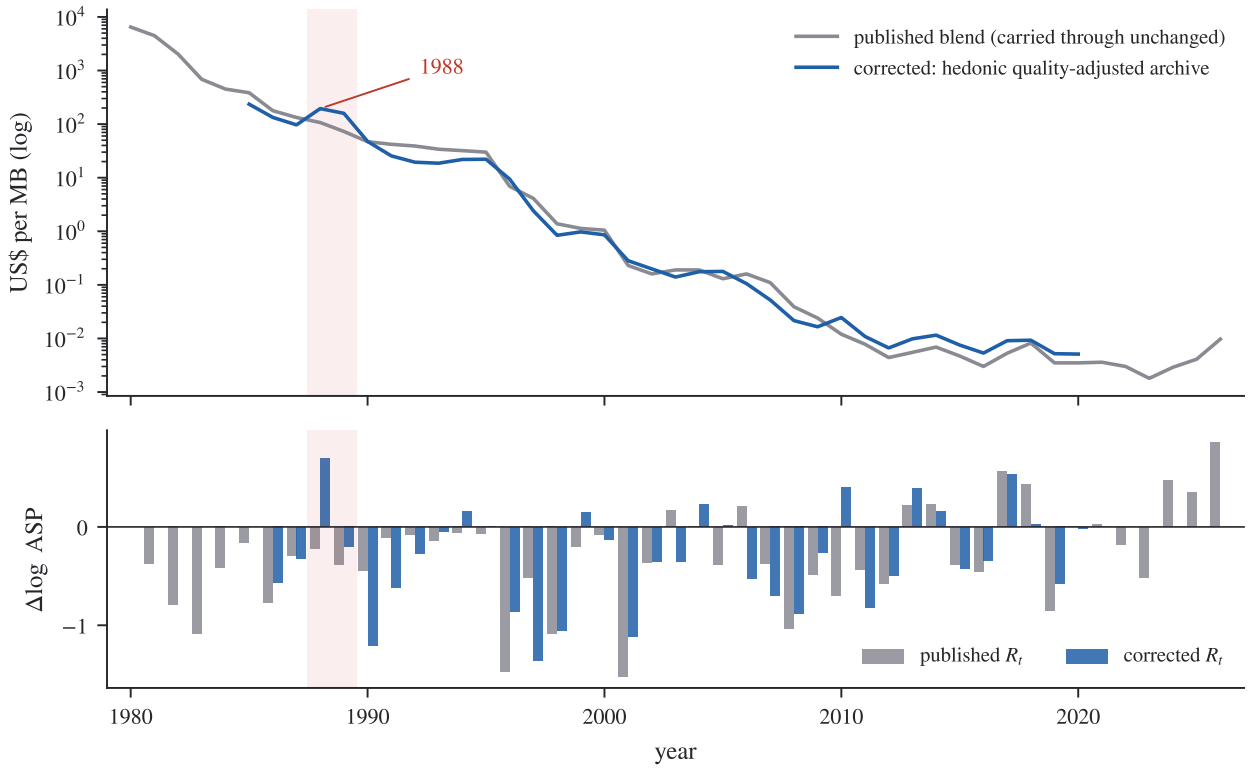


Figure 3: The published series and the corrected quality-adjusted series, published side by side; nothing is overwritten. *Top:* levels in US$ per MB. *Bottom:* the annual log changes that constitute the index. The two series agree nearly everywhere except the shaded 1988–89 episode, which the published table omits. Generated by `analysis/make_figures_v3.py`; the underlying panel is `data/indices-wide-v3.csv`.

7 RI-M: a monthly index, and why it does not work

7.1 Construction

RI-M is $R_t = \log P_t - \log P_{t-1}$ on the recovered archive, computed only between adjacent observed months. It spans 1985-06 to 2020-09 and yields 278 returns; 63 gaps totalling 150 months emit no return. It satisfies the Definition: rebuilding on the archive truncated at 2005-01 reproduces all 193 overlapping months with maximum absolute difference 0.0.

7.2 Why it fails

The construction is sound and the instrument is not. Two guards fire.

Check	Result
Months with no price change	21.6%
Longest flat run	10 months
Corr with IR213 import price index (monthly $\Delta \log$)	0.133
Corr with IQ213 export price index	0.029
Corr with semiconductor PPI	0.120

Table 2: RI-M against three independently constructed price indices.

The archive is a sequence of individual advertised retail quotes—*BYTE* and JDR magazine advertisements through the 1990s, NewEgg listings latterly—rather than a survey of transactions. The same quote is carried forward for months at a time ($504.83 from July 1988 to April 1989), and successive observations frequently switch to a different module. At annual frequency this averages out and the archive tracks other price measures at 0.45–0.60. At monthly frequency, which is the only frequency that would justify building it, it agrees with nothing. Figure 4 shows both failures at once: the staleness is visible in the return series itself, and the correlation with every independent measure collapses exactly when the aggregation window shrinks to the frequency the instrument exists for.

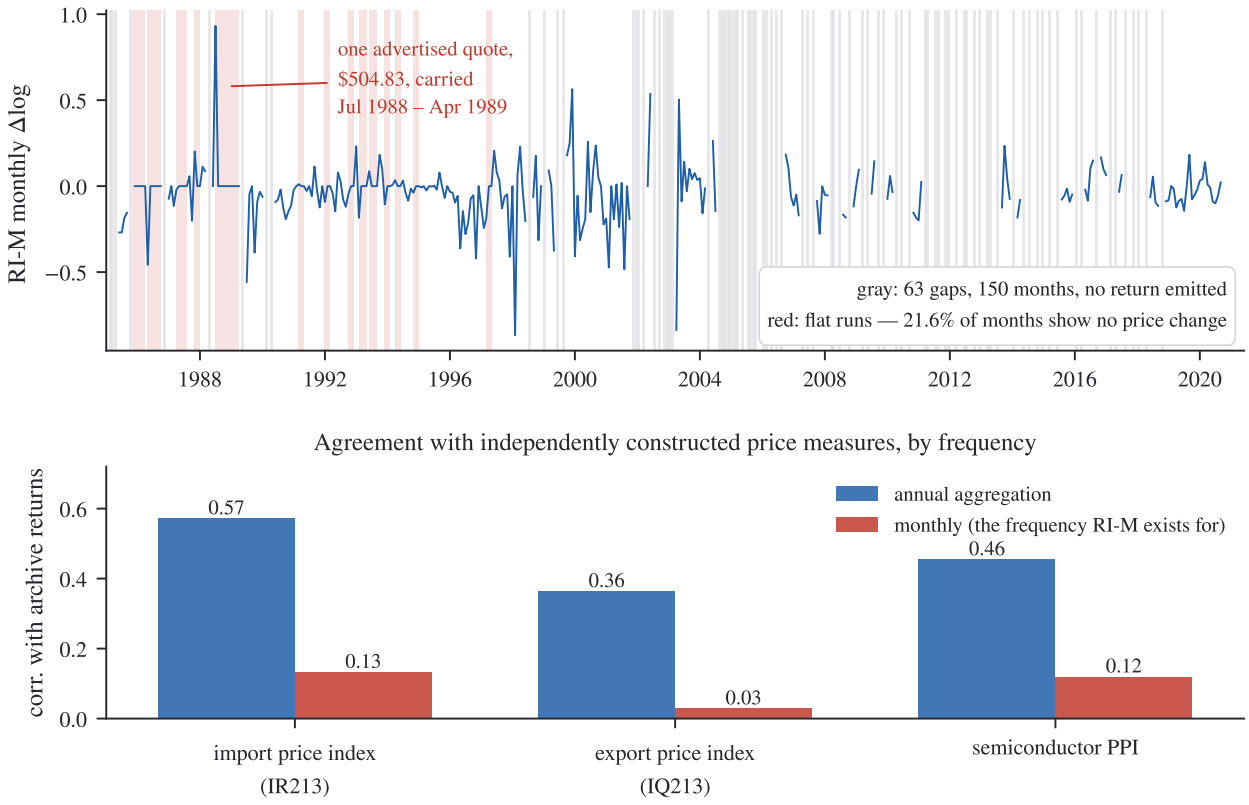


Figure 4: Why RI-M fails as an instrument. *Top:* the monthly return series; gray bands are the 63 gaps (150 months) across which no return is emitted, red bands are flat runs of three months or longer — carried advertised quotes, not markets that did not move. *Bottom:* correlation with three independently constructed price measures, computed at annual and at monthly frequency (Guard 7). The agreement that exists at annual frequency disappears at the frequency RI-M was built for. Generated by `analysis/make_figures_v3.py`; numbers match `analysis/out/FACTS-RIM.txt`.

7.3 The resulting verdict

RI-M leads nothing:

Test	n	best forward p	F ratio
RI-M → industrial production	278	0.622	1.26
RI-M → nonfarm payrolls	278	0.450	17.54
RI-M → real GDP (quarterly)	73	0.497	1.82

Peak lead correlation is 0.122, and every specification dies under the contemporaneous control. These are not marginal misses.

They are also not evidence. A null computed on an instrument that does not co-move with any independent measure of the same good measures the instrument, not the hypothesis.

The correct verdict is **untestable**—the same call the agricultural study made on cartel-administered potash prices, and for the same structural reason.

7.4 What this establishes

The monthly Ramification Index cannot be constructed from public or recoverable sources. This now follows from two independent directions:

1. No public price index measures DRAM. The Producer Price Index for semiconductor manufacturing correlates with the published DRAM series at $r = 0.186$ in annual log-differences—a value close to the 0.175 that identified two non-equivalent fertilizer series in the companion study. Two constructions within the BLS itself correlate at 0.286.

2. The one genuine archive of DRAM prices is too sparse and too stale to difference monthly, as above.

Licensed contract and spot ASP data at monthly frequency is therefore a precondition for testing the Ramification Index, not an enhancement to it.

8 Power: what the published panel can support

"No lead" and "not enough data to tell" are different findings, and neither Version 1 nor Version 2 distinguishes them. For an F-test of k restrictions at corrected level α, the smallest partial correlation detectable at 80% power is:

Panel	n	lags	α	min. detectable partial r
Published annual panel (v1, v2)	46	3	0.0167	**0.508**
RI-M quarterly	73	8	0.00625	0.518
RI-M monthly	278	12	0.00417	0.303
Hypothetical licensed monthly ASP	712	12	0.00417	0.192

Table 3: Minimum detectable effect by panel.

The published annual panel cannot detect a partial correlation below 0.508. That is a very large macroeconomic effect—larger than that claimed for almost any published leading indicator. A test with that little power cannot reject anything, and equally cannot establish anything at a corrected threshold. Figure 5 places each panel on the power curve for its own specification; the published panel sits far above the band in which real leading-indicator effects live, and only licensed monthly data reaches it.

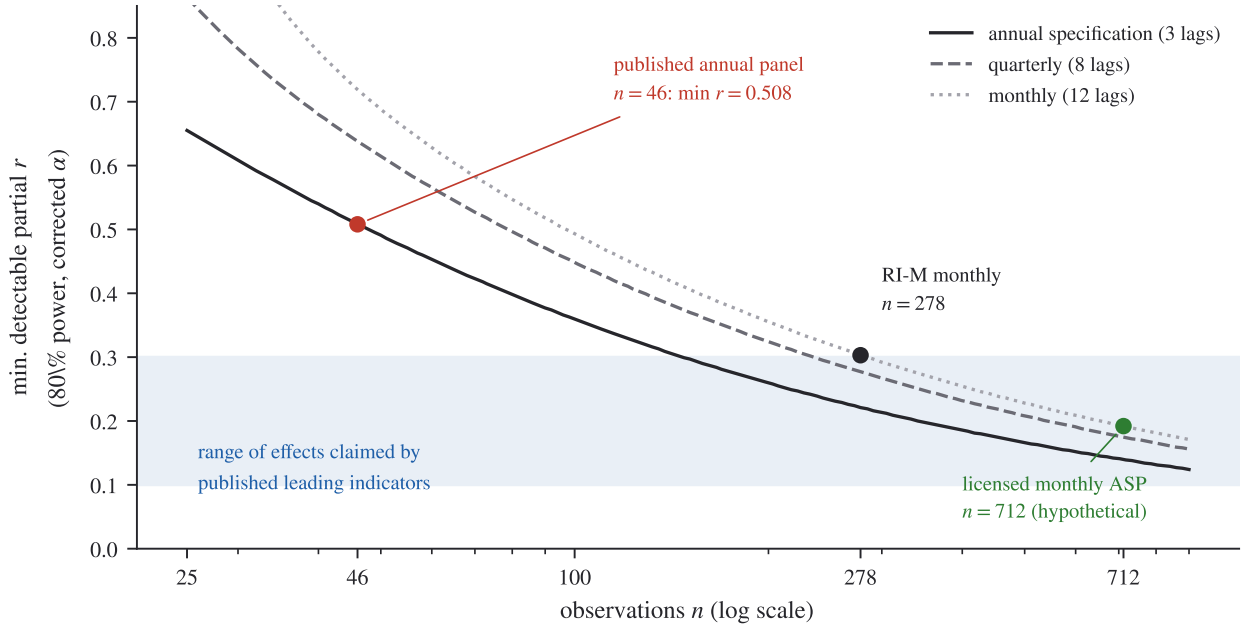


Figure 5: Minimum detectable partial correlation at 80% power against sample size, for the annual (3-lag), quarterly (8-lag) and monthly (12-lag) specifications at their multiple-comparison-corrected thresholds. The shaded band is the range of effect sizes claimed by published leading indicators. The published annual panel ($n = 46$) cannot see anything smaller than $r = 0.508$; a licensed monthly ASP series ($n \approx 712$) is the first panel whose floor sits inside the band. Generated by `analysis/make_figures_v3.py` using the same power computation as the protocol harness.

Run on the published panel directly, the index behaves as follows: `ram_index_R` → real GDP growth gives best forward $p = 0.0243$ at lag 2 against a corrected $\alpha = 0.0167$; the reverse direction is clean ($p = 0.499$, forward/reverse F ratio 5.78); and—notably —it *survives* the contemporaneous control with 98.9% of the F statistic retained. Whatever is in that series is anticipation rather than persistence.

8.1 Which observations carry the result

That last sentence holds on the full sample. It does not hold on most subsamples of it. Withholding years from the end of the panel and re-running the identical specification gives:

Sample	n	best fwd. p	clears α?	Guard 5
Full sample, as published	46	0.0243	no	SURVIVES
Drop 2026 (incomplete row)	45	0.0330	no	SURVIVES
Drop 2025–2026	44	0.0303	no	SURVIVES
Drop 2021–2026 (pre-RAMageddon)	40	**0.0847**	no	**DIES**
Drop 2020–2026 (pre-COVID)	39	**0.2984**	no	**DIES**
Drop 1981–1985	41	0.0159	yes	SURVIVES

Table 4: Endpoint sensitivity of the published Granger specification. $\alpha = 0.0167$ throughout.

Withholding the last six years takes p from 0.0243 to 0.0847 and flips the contemporaneous control from SURVIVES to DIES. Withholding the last seven takes it to 0.2984. The relationship is carried by 2020–2026: the COVID shock and the RAMageddon years—one of which is an incomplete row standing in a column of realised history (§4). Figure 6 walks the endpoint back one year at a time; the shape of that walk is the finding.

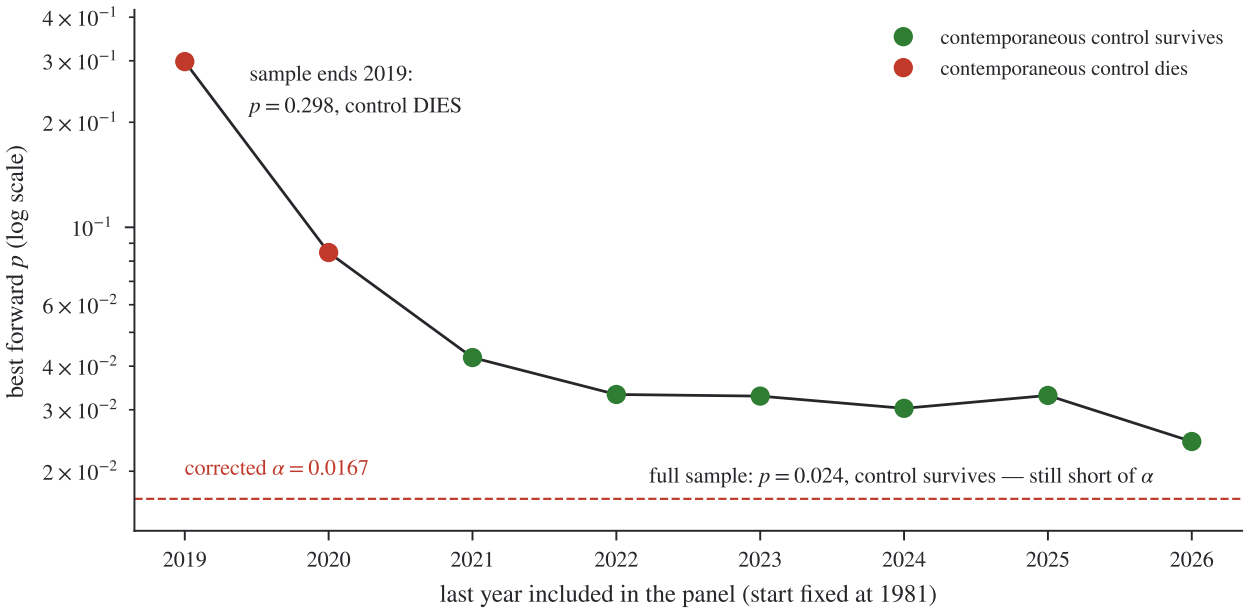


Figure 6: The endpoint walk. Each point re-estimates the identical annual specification (`ram_index_R` → real GDP growth, 3 lags) with the sample truncated at the year shown; the start is fixed at 1981. Green points survive the contemporaneous control (Guard 5), red points do not. The full-sample result sits closest to the corrected threshold and still short of it, and the entire approach toward significance occurs after 2019. Generated by `analysis/make_figures_v3.py`; values match Table 4.

The one specification that clears the corrected threshold drops 1981–1985 and does so at $p = 0.0159$ against $\alpha = 0.0167$. It is a sample choice made after seeing the result, and it is reported here only so that it cannot be reported later as a finding.

The corrected series behaves the same way, and slightly worse ($p = 0.1024$ on 1986–2020). But the *published* series also fails on that window ($p = 0.0423$) and dies under the contemporaneous control there too, so this is a property of the sample rather than an artifact of correcting it.

Remark 8.1 (What this audit does and does not reach)**.** This audit is run on the *annual* panel, which is the specification reproducible from the published data. Version 2's headline is the quarterly VAR(2), estimated on 1985Q1–2024Q4 with $F = 5.83$, $p = 0.004$.

Two things follow, and the second limits this paper. First, the quarterly specification ends in 2024Q4, so it cannot contain the 2025–2026 rows at all, and nothing here should be read as saying that it does. Second, and more importantly, the quarterly result is *not tested here* and cannot be: the underlying quarterly ASP series is not published, and this paper has been unable to reconstruct it from any available source. Whether the same endpoint dependence holds there—in particular whether 2020 carries it, as it does annually—is an open question, not a demonstrated result.

The claim made here is therefore narrower than it may appear: the annual specification's relationship with GDP is carried by its last six observations. Extending that to the quarterly headline would require the quarterly series, and until that series is available the honest position is that the headline is unaudited rather than undermined.

8.2 What this means for the claim

Version 2's Granger result is therefore restated rather than refuted, in a sharper sense than power alone would require. It is not merely that $n = 46$ is too small to resolve the effect. It is that the effect present in the full sample is contributed almost entirely by the last six observations, and is absent from the preceding forty.

That is consistent with two very different stories—a genuine relationship that only became detectable once DRAM grew macro-relevant, or a coincidence between two unusual recent episodes—and the panel cannot distinguish them. This is the fragility pattern the companion study documented: not a coding error, not an indefensible specification, but a result resting on a handful of recent observations and presented as a property of a 46-year series.

9 The protocol

Eight guards, seven inherited from the companion study, where each was written to kill a specific false positive, and one added here.

1. **Multiple-comparison correction.** Judge the best forward p-value against $\alpha = 0.05/\max$ lag. Reporting the minimum of N p-values against 0.05 is a selection procedure.
2. **Reverse direction, always.** A variable that responds to the cycle will often predict it slightly.
3. **Bidirectional is a verdict, not a tie-break.** When both directions survive, decide on F-statistic dominance; two p-values that have both underflowed cannot be compared.
4. **Point-in-time construction** of any derived signal (Section 5).

5. **Contemporaneous control.** A persistent predictor passes a standard Granger test when its lags merely proxy for its current value.
6. **Staleness.** A null on a series that does not move measures the freeze, not the hypothesis.
7. **Independent-source replication.** A positive result must survive substitution of an independently constructed measure of the same quantity.
8. **Definitional overlap** (new). A test between an aggregate and a series inside it measures an accounting identity. Industrial production for semiconductors against total industrial production returns $p = 1.9 \times 10^{-8}$, the largest statistic obtained anywhere in this work, and predicts nothing.

Additionally: year-over-year changes on higher-frequency data use overlapping windows, inducing MA($k-1$) residual structure and inflating test statistics—the classic overlapping-data problem [Hansen & Hodrick 1980, Newey & West 1987]. Every year-over-year verdict in the companion study reversed under log-differences.

Against this protocol, Version 2 passes Guards 1, 2 and 3—its reverse test is reported ($F = 2.14$, $p = 0.121$) and its F ratio of 2.72 clears the dominance threshold, which is uncommon practice and worth stating. Guards 4 through 7 have not been run on it, and Guard 5 has never been run on the quarterly specification at all.

Remark 9.1 (On placebos, including one of this paper's own)**.** Version 2 reports NAND flash as a failing placebo. On Version 2's own mechanism this is not a valid placebo: NAND is bought by the same capacity-planning firms, for the same deployments, on the same planning horizon, and should therefore lead if DRAM does. A placebo must fail the mechanism, not merely be a different product.

An earlier draft of this paper proposed passive components—capacitors and resistors, bought on short lead times against realised orders—as the appropriate substitutes, and reported that they return clean nulls. They do at monthly frequency. At the annual frequency of the headline specification they do not, and the claim was wrong:

Series → real GDP (annual, 3 lags)	n	best fwd. p	Guard 5
Capacitor PPI (proposed placebo)	43	**0.00195**	SURVIVES
Storage-device PPI	34	0.0130	SURVIVES
DRAM, published `ram_index_R`	46	0.0243	SURVIVES
Resistor PPI	43	0.2010	dies
Semiconductor PPI	47	0.2493	dies
PPI all commodities	47	0.1873	dies
Crude oil (WTI)	47	0.4575	dies

Table 5: Annual lead tests against real GDP growth, identical specification. $\alpha = 0.0167$.

The capacitor PPI—a component bought against realised orders, with no forward-commitment mechanism to speak of—produces a *stronger* result than DRAM: $p = 0.00195$, a clean reverse

direction (forward/reverse F ratio 39.3), and 95.8% of the F statistic retained under the contemporaneous control. It clears the corrected threshold that DRAM does not.

Three of eight price series "lead" GDP on this panel. That is the more informative finding, and it is not a finding about capacitors. It is what an $n \approx 45$ annual panel does when several correlated macro-sensitive series are each given three lags and a corrected threshold that only just binds. It should be read as strengthening §8 rather than as a discovery about passive components, and it is the reason this paper reports no verdict it has not run at the frequency it is claiming it for. Figure 7 shows the full panel at a glance.

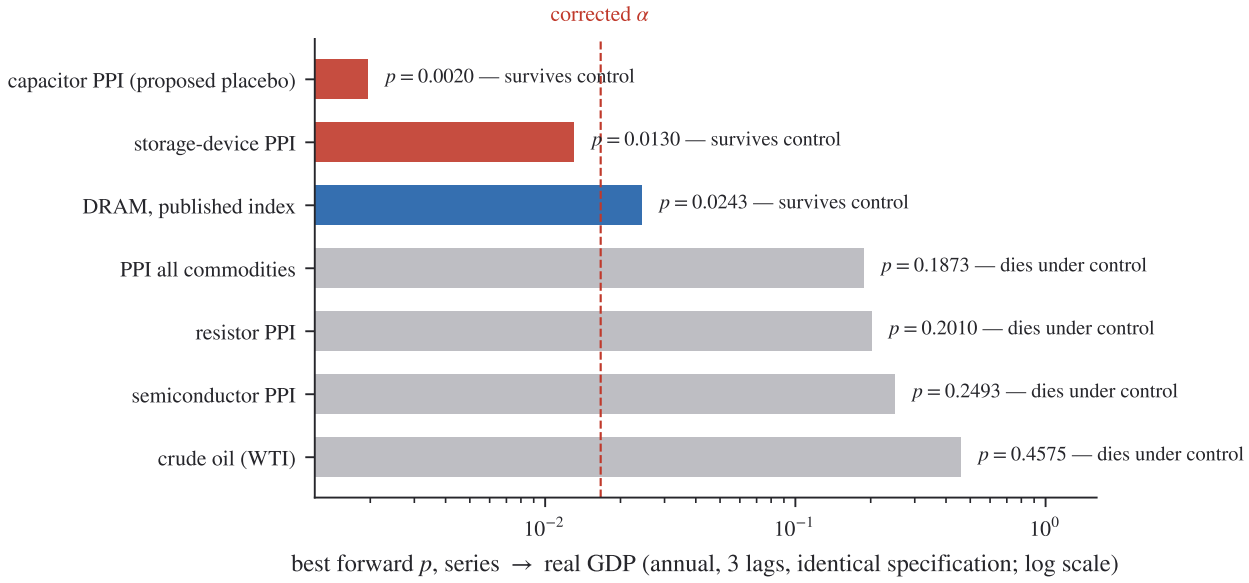


Figure 7: Every series in Table 5, identical specification, sorted by best forward p-value (log scale). Red bars survive the contemporaneous control; the blue bar is the published DRAM index; gray bars die under the control. A component with no forward-commitment mechanism outperforms DRAM on DRAM's own test, which is a statement about what this panel can distinguish, not about capacitors. Generated by `analysis/make_figures_v3.py`.

10 A scope condition

The companion study tested six candidate leading indicators across the agricultural input complex and found none, then explained why in structural terms. DRAM prices lead because memory is bought by a few hundred capacity-planning firms with forecasting functions and capital-market access, ahead of deploying compute: the purchase is a forward expectation expressed as a price. Agricultural inputs are bought by roughly two million atomised price-takers on an agronomic calendar, funded largely by last season's realised cash. No agent in that system has both the incentive and the scale to express a forward expectation as a price.

This converts the Ramification Index from a single working example into a class of indicator with a stated and falsifiable scope condition: *concentrated buyers, discretionary*

forward commitment, and a fast, elastic, private-market price. Agriculture is then a tested and failed boundary case rather than an unexamined gap. Farm machinery—financed capital expenditure, the closest structural analogue to DRAM—leads nothing either, which sharpens the condition to concentration rather than merely capital intensity. Figure 8 lays the tested cases out against the condition.

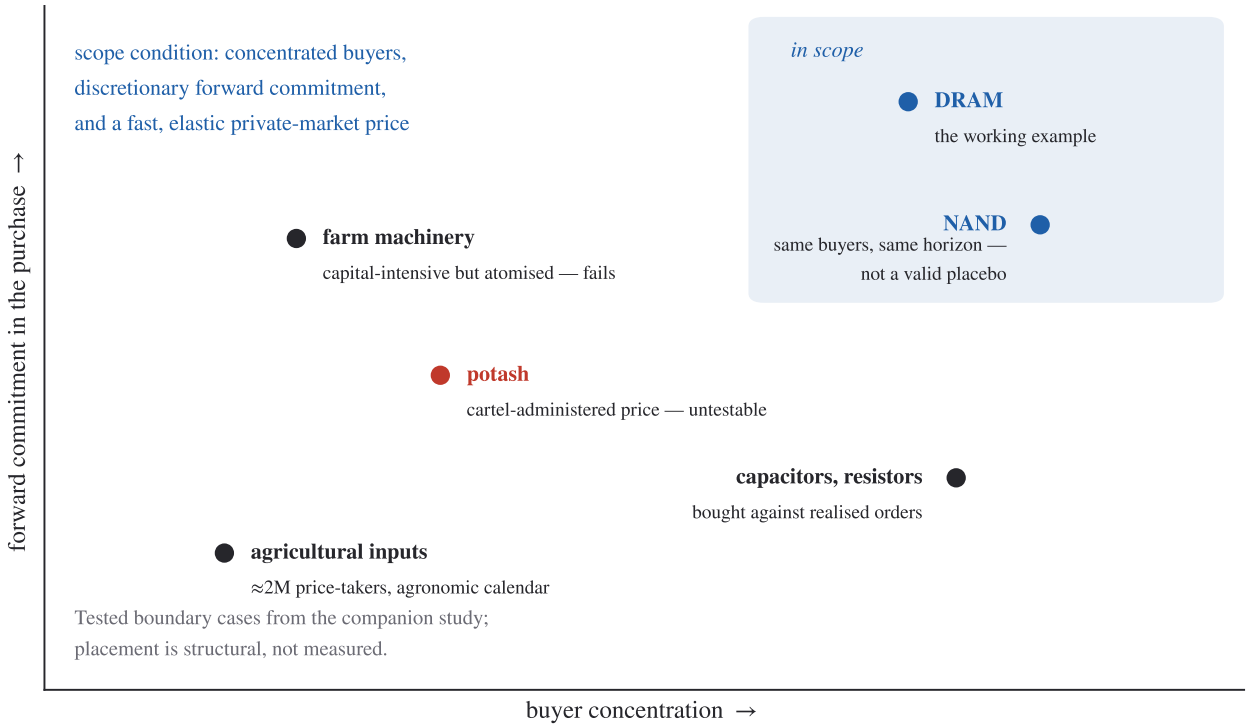


Figure 8: The scope condition and its tested boundary cases. Placement is structural — who buys, on what horizon, at what kind of price — not measured; the diagram carries no estimated quantities. NAND sits inside the scope region, which is exactly why it cannot serve as a placebo (Section 9); potash is marked untestable because its price is cartel-administered, the verdict the companion study reached and RI-M inherits.

The present work sharpens it further, and at this paper's own expense. Neither the broad semiconductor price complex nor semiconductor output satisfies the condition: both fail the contemporaneous control, and generic industrial production leads employment at $p = 1.7 \times 10^{-20}$, which is what an identity between a leading and a lagging aggregate looks like. Whatever the Ramification Index measures is narrower than "semiconductors," and the paper should say so.

11 What is withdrawn and what is restated

Withdrawn:

- The claim that the 2025–2026 episode is "without historical precedent" in the 46-year series. The series omits the 1988 shortage (Section 4); the comparison cannot be made on this data.
- The description of the price series as a McCallum–TrendForce chain-link (Section 3). It is a rounded three-source retail blend.
- Any inferential reading of the bifurcated sub-indices. `ram_commodity_R`, `ram_ai_R` and `hbm_weight` are populated for three annual observations, one of which is a forecast. No lead test is estimable; the bifurcation stands as a descriptive regime claim only.
- The characterisation of NAND flash as a placebo test.

Restated:

- The Granger result. It is not refuted—on the full sample it survives the contemporaneous control and its direction is unambiguous—but it does not clear a corrected threshold, it is carried entirely by 2020–2026 (§8.1), and the quarterly estimation cannot presently be reproduced because the underlying series is not published. Any restatement should report the endpoint sensitivity alongside the headline p-value.
- The recession-coverage and correlation results, which stand as computed but rest on a series with a documented omission.

Retained: the structural argument for DRAM (oligopoly, forward commitment, elastic private-market pricing); the reverse-direction discipline; and the historical and institutional material on cartel behaviour.

12 Data and code availability

The recovered McCallum workbook, the eight-guard harness with its 20 regression tests, the index construction with its point-in-time proof, and verbatim run output for every figure in this paper are available at https://github.com/fromknowware/bifurcation-memory-index under `analysis/`. All macroeconomic series are fetched keylessly from FRED and cached, so the battery is reproducible offline.

Every figure in this paper traces to `analysis/out/FACTS.txt`, `FACTS-RIM.txt`, or `FACTS-CORRECTED.tx`. The plotted figures are generated by `analysis/make_figures_v3.py`, which recomputes every measured quantity from the same modules the batteries use rather than reading numbers from this paper; the one exception, the scope diagram, is conceptual and carries no measured quantities. Where a number here and a number there disagree, the run output is correct and this paper is wrong.

The corrected panel is `data/indices-wide-v3.csv`. It carries every published column through unchanged—so all Version 1 and Version 2 figures remain reproducible from it—and adds `ram_usd_per_mb_qa`, `ram_index_R_qa`, `ram_qa_n_obs`, a per-year `ram_provenance` string, and a `row_status` flag marking incomplete years.

13 Conclusion

The Ramification Index may well lead the business cycle. The argument for why it should is structural, specific, and now equipped with a scope condition that makes it falsifiable. What this version establishes is that the published data cannot currently settle the question: the primary source had to be recovered from a web archive, the series omits one of the two largest supply shocks it covers, the construction is not point-in-time-verifiable, and the panel is two orders of magnitude too small in the dimension that matters.

The sharpest version of the problem is Table 4. A relationship that holds on 46 observations and vanishes on the first 40 of them is not yet a property of the series; it is a property of the last six years, one of which has not finished happening. Whether those six years are the moment DRAM became macro-relevant or merely two unusual episodes in a row is exactly the question the data cannot answer, and it is the question worth answering.

None of that is a refutation. It is a statement about what has been demonstrated versus what has been asserted, and the distinction is the contribution. The protocol that produced it converted four apparently publishable findings into zero in agriculture; applied here it converts one asserted finding into an open question and two documentation claims into corrections. That is the expected yield of a good protocol, and the correct response is to acquire the data the question requires rather than to lower the standard until the existing data clears it.

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